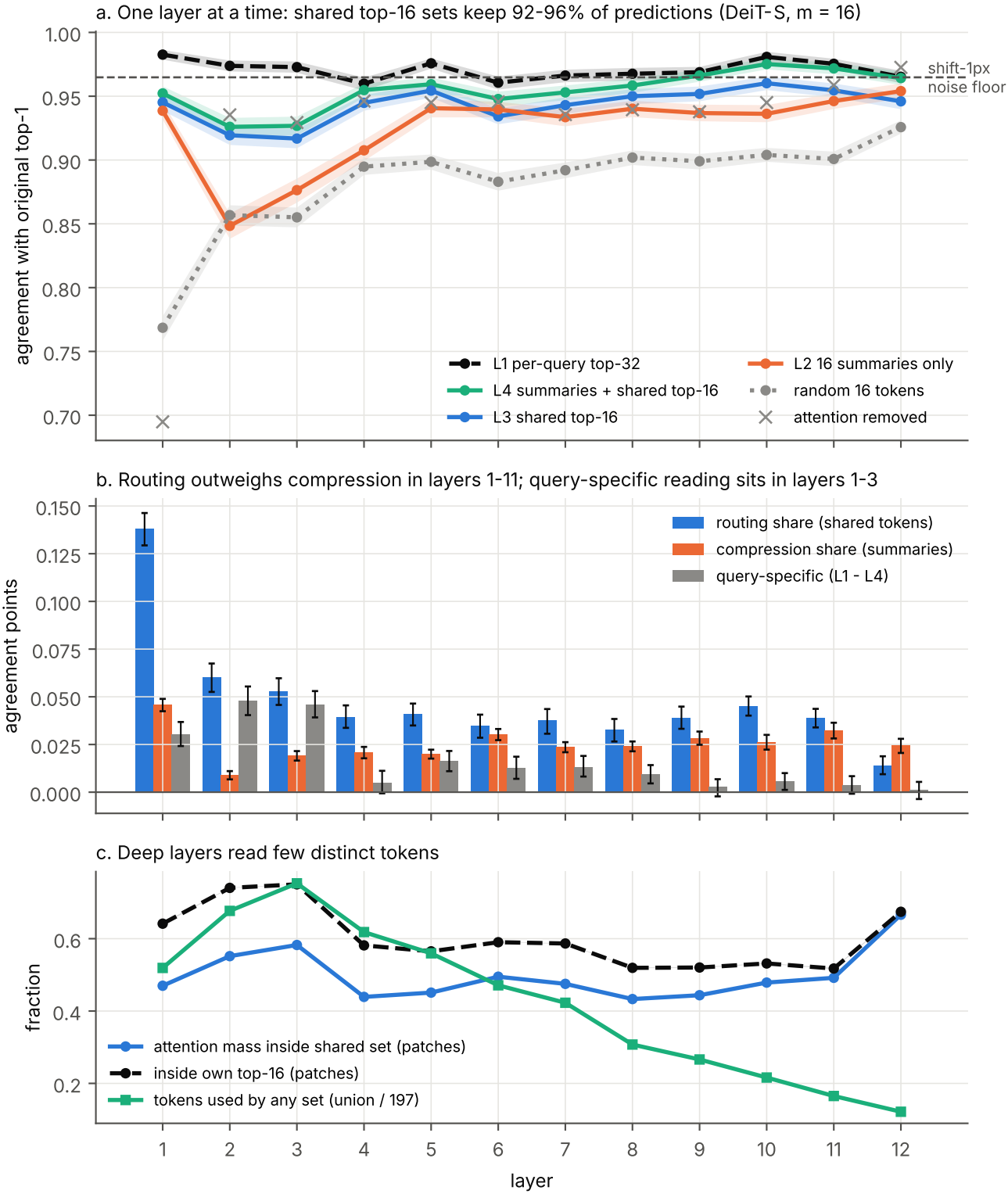
a. One layer at a time: shared top-16 sets keep 92-96% of predictions (DeiT-S, m = 16)
agreement with original top-1
1.00
0.95
0.90
0.85
0.80
0.75
0.70
shift-1px
noise floor
L1 per-query top-32
L4 summaries + shared top-16
L3 shared top-16
L2 16 summaries only
random 16 tokens
attention removed
b. Routing outweighs compression in layers 1-11; query-specific reading sits in layers 1-3
agreement points
0.150
0.125
0.100
0.075
0.050
0.025
0.000
routing share (shared tokens)
compression share (summaries)
query-specific (L1 - L4)
c. Deep layers read few distinct tokens
fraction
0.6
0.4
0.2
attention mass inside shared set (patches)
inside own top-16 (patches)
tokens used by any set (union / 197)
1
2
3
4
5
6
7
8
9
10
11
12
layer